

THERMOPLASTIC

A thermoplastic, or thermoset plastic, is any plastic polymer material that becomes pliable or mouldable at a certain elevated temperature and solidifies upon cooling.

Most thermoplastics have a high molecular weight. The polymer chains associate by intermolecular forces, which weaken rapidly with increased temperature, yielding a viscous liquid. In this state, thermoplastics may be reshaped and are typically used to produce parts by various polymer processing techniques such as injection moulding, compression moulding, , and extrusion. Thermoplastics differ from thermosetting polymers (or “thermosets”), which form irreversible chemical bonds during the curing process. Thermosets do not melt when heated, but typically decompose and do not reform upon cooling.

5 Benefits of Thermoplastic Components

- Chemical resistance — Thermoplastics are often the material of choice for seals and valves, as properly chosen materials are slow to degrade or lose performance when in contact with gases or fluids. This is a critically important differentiator in can’t-fail applications or extreme environments, such as oil and gas pipelines, medical implants, or aerospace. Because the gas or fluid is slow to (or simply can’t) permeate the polymer material, components won’t rust, corrode, or degrade like other materials.
- Strength-to-weight ratio — Particularly for aerospace and automotive applications, in which weight is an important consideration, thermoplastics are the preferred choice over metal parts because they are both strong and lightweight.
- Durability — Because of their combination of abrasion, temperature, pressure, and fatigue resistance, engineered thermoplastics are unparalleled in durability.
- Cost — Thermoplastics are easier to manufacture relative to metals, and net-shape or near to net-shape articles can be produced more readily at lower total cost. Thermoplastics are easily melted at moderate temperatures and more readily formed into a finished or semi-finished part, avoiding or reducing the complexity of secondary operations, thus reducing production costs.
- Sustainability — Plastics get a bad rap, but responsible manufacturers are working hard to make them more sustainable. At CDI, we are always looking at ways to recycle and reuse our thermoplastic scrap stream, with a goal of keeping plastic waste out of landfills and out of the environment. We also operate on 100% renewable energy at our Houston facility.

Examples of thermoplastics

- Polyethylene
- Polypropylene
- Polyvinyl chloride
- Polystyrene
- Teflon.

THERMOSETTING PLASTIC

A Thermosetting plastic is a polymer that is irreversibly hardened by heat. Thermosetting plastics are also known as thermosets, thermosetting polymers, or thermosetting resins. The starting material for a thermoset is a liquid or soft solid. Heat provides energy for covalent bond formation, cross-linking the polymer subunits and curing/hardening the plastic. Sometimes the heat is applied externally, but it may come from the chemical reaction of mixing ingredients. Adding pressure, a catalyst, or a hardener can increase the curing rate. Once cured, a thermosetting plastic cannot be re-melted, so it is formed into its final shape by injection moulding, extrusion moulding, compression moulding, or spin casting.

Thermoset plastics provide a wide range of benefits:

- Able to be moulded with different tolerances
- Allows for flexible product designs
- Improved structural integrity through variable wall thicknesses
- Typically cheaper than components fabricated from metals
- Superb electrical insulation properties
- Excellent heat resistance at high temperatures
- Corrosion resistant
- Strong dimensional stability
- Low thermal conductivity
- Cheaper setup and tooling costs than with thermoplastics
- High strength-to-weight ratio
- Water resistant
- Wide range of colours and surface finishes

Thermosetting Plastic Examples

Many plastics encountered in daily life are thermosetting plastics. Examples include:

- Bakelite (phenolic)
- Cyanate esters
- Epoxy resin
- Fiberglass (a fibre-reinforced thermoset)
- Melamine
- Polyester resin
- Polyurethane
- Silicone resin
- Vinyl esters
- Vulcanized rubber.

Difference Between Thermosetting Plastic and Thermoplastic

Heat makes a thermosetting plastic irreversibly rigid, but it makes a thermoplastic moldable or pliable. A thermoplastic then hardens again upon cooling. Thermosetting plastics tend to be stronger than thermoplastics because of internal cross-linking via covalent bonds. For the same reason, thermosetting plastics tend to have higher corrosion resistance and hardness. On the other hand, thermosets are more likely to permanently deform under a load and are more brittle than thermoplastics. Thermosets cannot be reshaped, but they are perfect for high-temperature applications, including electronics and appliances. Thermoplastics can be remoulded and recycled. Their strength, flexibility, and shrink-resistance make them suitable for high-stress parts and plastics bags and containers.

What Is Plastic Compounding?

While plastics continue to be one of the most used materials in homes and businesses daily, their methods of use constantly change. This is especially true also in the way we produce and engineer plastics. The experts at Plastic Service Centre see plastics as a way to continue combining existing plastic with new natural resources. One of these plastic manufacturing methods is called “plastic compounding.”

Compounding is a processing of plastics that changes the existing characteristics of engineered thermoplastics. From changing thermal components to the overall aesthetic and dimensions of plastics, compounding melts down a blend of plastic to solidify into a new product.

How Does Plastic Compounding Actually Work?

Thermoplastic compounding makes the use of plastic polymers easier and more efficient. The process of compounding will essentially cause the natural materials within the plastic to behave differently, especially when combined with other materials. While in a compounding extrusion machine, the plastic is heated, melted, and propelled forward into the barrel. Here, the additives that change the characteristics of the plastics are added along the length of the barrel to mix and alter the polymer base. When the plastics are melted through and combined with the additives, it is then time to use the new plastic resins to make end products.

Moulding Techniques of plastics

- Compression Moulding.
- Injection Moulding.
- Transfer Moulding.
- Extrusion Moulding.

1.Compression Moulding

This method involves the raw liquid plastic being poured into a heated mould and then being compressed together to form the desired shape. The high temperature of the entire process

ensures good strength in the final product. The process is finished off by cooling the liquid plastic so that it keeps its form before being trimmed and removed from the mould.

Common Uses of Compression Moulding

The most effective use for this method is when you want to make plastic replacement parts for broken metal parts. The reason for this is because the high-temperature method makes for a very strong and durable end product. Even though it is plastic, it is a strong and low-cost replacement for the metal parts.

2. Injection Moulding

Injection Moulding is fairly similar to extrusion moulding. The difference here is that with injection moulding the melted plastic is injected directly into a custom mould. The injection is under high pressure so that the mould is filled and a solid part is made. As with the other methods, after the mould is filled, the plastic is cooled as to keep its new shape and then the mould is opened. This can be compared to a Jello mould when the mould is filled and then cooled

Common Uses for Injection Moulding

This is a common method to produce a high volume of plastic parts like car parts or even parts for surgical applications. Products can also be made with increased flexibility to suit the needs of designers or engineers.

3. Transfer Moulding

Transfer moulding process combines the principle of compression and transfer of the polymer charge. Transfer moulding is similar to compression moulding except that instead of the moulding material being pressurized in the cavity, it is pressurized in a separate chamber and then forced through an opening and into a closed mould. The advantages of transfer moulding are that the preheating of the material and injection through a narrow orifice improves the temperature distribution in the material and accelerates the crosslinking reaction. As a result the cycle times are reduced and there is less distortion of the moulding. The improved flow of the material also means that more intricate or complicated shapes can be produced. Transfer moulding process combines the principle of compression and transfer of the polymer charge.

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Applications

This process is widely used to encapsulate items such as integrated circuits, plugs, connectors, pins, coils, and studs. It is suitable for moulding with ceramic or metallic inserts which are placed in the mold cavity. When the heated polymer fills the mould it forms bonding with the insert surface. Transfer Moulding is also used for manufacturing radio and television cabinets and car body shells.

4.Extrusion Moulding

With extrusion moulding, hot melted plastic is pressed through a shaped hole to create a lengthy shaped plastic part. This customizable shape that the liquid plastic gets pressed through is called a die. This die is custom made for the particular outcome that is desired. It is almost like pressing dough through a press to make shaped cookies.

The other forms of plastic moulding also use extrusion so get the raw liquid into the moulds, the difference here is that other methods use the moulds to make the desired shape and here the extrusion itself is making the shape with the use of the die's shape.

Common Uses of Extrusion Moulding

When using this method your outcome product will continuously have the same shape along the length of it. These can be things like straws or PVC pipes. These types of parts can be made at very high volumes because it can just keep producing the same shape without end.

In comparison, this type of moulding is low cost because the equipment is fairly simple and can have high productivity. The downside to this method is that you are very boxed in when it comes to the variety of parts you can make.